A Few More Series

1. Do the following series converge or diverge? State the test you used. For convergent series with negative terms, indicate if they converge conditionally or absolutely. If appropriate, find the sum.
   (a) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{k+4}{3k-1}\right)^k$
   (b) $\displaystyle\sum_{k=2}^{\infty}\frac{k^3-3k}{1+\ln k}$
   (c) $\displaystyle\sum_{k=3}^{\infty}\frac{1}{k}-\frac{1}{k+1}$
   (d) $\displaystyle\sum_{k=1}^{\infty}k\frac{3^k}{4^k}$
   (e) $\displaystyle\sum_{k=0}^{\infty}\frac{(-1)^k}{2\sqrt{k}+3}$
   (f) $\displaystyle\sum_{k=2}^{\infty}\frac{3}{k\ln k}$
2. What is the difference between absolute and conditional convergence?
3. Suppose $a_k > 0$ and $\displaystyle\lim_{k\to\infty}\frac{a_k}{1/k}=\frac{1}{2}$. Does $\displaystyle\sum_{k=0}^{\infty}a_k$ converge or diverge? Why?
4. If $\displaystyle f(x)=\sum_{k=0}^{\infty}\frac{5k}{k!}x^k$, what is $f''(0)$?
5. Indicate if the following are always true or may be false.
   (a) _______ If $\sum_{k=0}^{\infty}a_kx^k$ converges when $x=4$, then it must converge when $x=3$.
   (b) _______ If $\lim_{k\to\infty}a_k=0$, then $\sum_{k=0}^{\infty}a_k$ converges.